Yi Nathen Qing

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SUMMARY OF QUALIFICATIONS

- Passionate undergraduate researcher interested in synthetic biology, biocomputation, protein design, and tissue engineering.
- Experienced leader from awards-winning high school FRC robotics team and college iGEM competitions.
- **Technical skills**: experience in Java, Python(numpy, matplotlib, seaborn, scipy, tkinter...), MATLAB, CAD, Bash, and COMSOL.
- **Wet-lab skills**: DNA purification, E.coli protein purification, E.coli and mammalian cell culturing, colony PCR, GGA, double photon microscopy, extrusion-based bio-printing, UV-vis spectroscopy.

EDUCATION

University of Washington, School of Engineering | Seattle, WA **EXPECTED GRADUATION: June 20, 2027**

- Major: Bioengineering: Data Science Option; Dean's List
- Related Coursework: Honors Chemistry, Physics, and Biology Series; Matrix Algebra, Differential Equations, Databases, Algorithms, Signal Processing, Fluids and Biotransport, Organic Chemistry, Machine Learning, Solid and Gel Mechanics.

American Heritage School, High School Diploma | Delray Beach, FL **June 10, 2023**

- GPA: 5.11 (Cumulative)
- National Honors Society, AP Scholar, Science Fair Research Board Excellence; Pre-med track, Engineering track

PROFESSIONAL EXPERIENCE

Undergraduate Research Intern | UW Institute of Protein Design | Seattle, WA **Nov 2024 - Present**

- Designing and testing de-novo light harvesting complexes under the guidance of Avi Swartz as an undergraduate research intern.
- Writing in-silico parametric design pipeline for photosynthetic fiber design and running wet-lab validation, purification, and assays.

Undergraduate Researcher | UW Institute of Protein Design JUPITER Cohort | Seattle, WA **Sept 2024 - June 2025**

- Applied to and was accepted into the IPD JUPITER Cohort program.
- Designed and tested phosphorylation-inducible heterodimers with Kathryn Shelley and Cullen Demakis for JUPITER.

Undergraduate Researcher | UW Medicine Zheng Lab | Seattle, WA **Sept 2023 - Dec 2024**

- Accepted as a first-year undergraduate to work with Dr. Sam Reyner in modeling microscopic functional intravascular topologies.

Research Intern | Oregon Health and Science University Nakayama Lab | Portland, OR **June 2024 - Sept 2024**

- Studied under the guidance of Dr. Karina Nakayama and members on biomaterial fabrication and statistical data analysis.
- Engineered open source Voron 3D printer into a bioprinter.
- Developed and publishing methods for autonomous fabrication of shear-induced scaffolding material for tissue regeneration.

Student Mentor/Programmer | American Heritage School | Delray Beach, FL **July 2021 - Aug 2021**

- Mentored as a senior student instructor.
- Guided students on FTC and FRC robot programming and engineered chassis for software testing.
- Programmed software for HPCC competition project: autonomous security patrol robot with facial recognition and identification.

ADDITIONAL ACTIVITIES

Founding Member | Adam Biotech | Seattle, WA **April 2026 - Present**

- Interviewed and selected for R&D team as founding member of student-led startup company manufacturing bio-printing equipment.
- Advanced to sweet 16 in UW Dempsey Business Competition and launching products in June.

Undergraduate iGEM Researcher | University of Washington | Seattle, WA **Sept 2023 - Present**

- Chosen as a member of the University of Washington iGEM Research and Wetlab development team and won gold medal in the 2023-2024 season working on Might Moeities project.
- 2024-2025 worked on designing and testing de-novo competitive inhibitors for exoA excretion proteins.

Yale Summer Session | Yale University | New Haven, CT **July 2023 - August 2023**

- Selected for and attended Yale Summer Session: took college-level chemistry and visual arts courses ending with As and received a letter of recommendation from Dr.Ganapathisubramanian.

High School Scientific Research | American Heritage School | Delray Beach, FL **August 2019 - May 2023**

- Wrote a review paper on current issues regarding current bottlenecks in in-vitro meat product under the guidance of Dr.Thompson.
- Programmed ML based software for classifying myosatellite tissue fitness through phenotypical features.

High School Robotics Competition Team | American Heritage School | Delray Beach, FL **August 2019 - May 2023**

- Lead Programmer of High school FRC and FTC teams. Used Java and C++ extensively during machine vision development.
- Entered FRC Worlds quarterfinals during 2022-2023 school year.